

ELEMENTS OF MECHANICAL ENGINEERING DESIGN

I Year II Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME06	ESC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OVER VIEW: The purpose of this course is to introduce the fundamental principles of Mechanical Engineering Design which is very important and relevant in the context of today's engineering professionals.								
COURSE OUTCOMES: The course should enable the students to: 1. Understand the working, usage and applications of different measuring instruments 2. Select suitable material fit for the purpose. 3. Identify design principles from an engineering perspective. 4. Devise visual design using conceptualization techniques 5. Choose an appropriate reverse engineering technique for the component design.								
UNIT-I	MEASUREMENTS							
Vernier caliper, micrometer and screw gauge – types, usage and applications; Transducers – Force, Torque, Strain, Temperature and Pressure; Accuracy, precision and resolution; measurement errors; Introduction to fits and Tolerances. Practice / Demonstration: Measurement of length, height and diameter; Measurement of Force, Temperature and Pressure								
UNIT-II	MATERIAL SELECTION							
Commonly used materials, standard sizes and shapes, Fit for purpose, The role of materials selection in design, Modulus-Density, Strength-Density, Material selection process, Introduction to 3D printer materials; Case studies and Brain storming.								
UNIT-III	MECHANICAL DESIGN FUNDAMENTALS							
Fundamental principles – Laws of nature, Simplicity Vs Complexity, Golden rectangle, Functional independence, Reciprocity, Self principles, Stability, Symmetry, Triangulate for stiffness, Load paths, Avoid bending stresses, Managing of friction, Preload, Centers of action, Exact constraints and controlled compliance; Case studies and Brain storming.								
UNIT-IV	DESIGN THINKING							
Design Conceptualization and Philosophy, Original, Adaptive, Variant and Re-Design, Innovation, Types of innovation, Conceptualization techniques: Idea generation, brain storming, Mind maps, SCAMPER, TRIZ, Biomimicry, Shape mimicry, SWOT analysis; Case studies and Brain storming.								
UNIT-V	REVERSE ENGINEERING							
Objectives and common application fields, Existing technologies, Contact systems, Non-contact systems, Introduction to Prototyping – 3D Scanning, 3D Printing, Laser Cutting, Plasma Cutting, Water Cutting, Introduction to CNC Machines. Practice / Demonstration on: 3D Printer, Laser Cutting machine, CNC Lathe and Milling machine; Simple parts - Prototype preparation.								

Reference Books:

1. Anand K Bewoor and Vinay A Kulkarni, Metrology & Measurements, McGraw Hill Education, 2017.
2. R. K. Jain, Mechanical and Industrial Measurements, 12th edition, Khanna Publishers, 2011.
3. Michael F. Ashby, Materials Selection in Mechanical Design, 5th Edition, Butterworth-Heinemann, 2016.
4. Ullman. D.G, The Mechanical Design Process, McGraw-Hill, 1997.
5. Otto. K and Wood, K, Product Design, Pearson Education, 2001.
6. Daniel Ling, Complete Design Thinking Guide for Successful Professionals, Create Space Independent Publishing, 2015.
7. Jimmy Jain, Design Thinking for Startups: A Handbook for Readers and Workbook for Practitioners, Notion Press, 2018.